Task 1: what the data looks like and the gap I found

1. What the Data Looks Like

- **Data Scale & Quality:** Clean, massive, continuous time-series dataset (2.55M+ samples at 2000 Hz) with 0 missing values and 0 duplicate frames.
- **Signal Volatility:** Microvolt-level signal featuring high kurtosis (up to ~222) and frequent high-amplitude contraction spikes (8%–25% outliers).
- **Feature Relationships:** High correlation (r > 0.85) among amplitude features (MAV, RMS, WL), while Zero Crossing (ZC) provides independent frequency data.
- **Manifold Complexity:** Non-linear distribution where linear methods (PCA) fail to separate gestures, but non-linear projections (UMAP) isolate Rest from dynamic movements.

2. Identified Gaps & Modeling Challenges

- **Class Distribution Gap:** Severe "Rest" bias where Class 0 occupies 50%–70% of the entire timeline, creating an artificial ~70% baseline accuracy trap.
- **Spatial Resolution Gap:** High single-channel ambiguity where multiple distinct gestures (e.g., 2, 4, 8, 10, 12, 15) exhibit identical median and IQR amplitude ranges.
- **Feature Redundancy Gap:** High multicollinearity (r > 0.90) across time-domain features on neighboring channels, which can destabilize traditional linear models.
- **Pipeline Mitigation Gap:** Bridging these hurdles requires robust scaling (to preserve real burst spikes), multi-channel spatial/temporal models (GNNs), and imbalance-aware validation (Macro-F1).